

Dharun Saravanakumar

+91 6381814730 | ddharun887@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

B.Tech in Electronics and Communication Engineering

August 2023 – Present

Indian Institute of Information Technology, Sri City — GPA: 8.4/10.0

Chittoor, AP

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Embedded Systems, IoT, OOP, Computer Networks.

TECHNICAL SKILLS

Programming Languages	C/C++, Python, JavaScript (ES6+), SQL, MATLAB, Verilog
Frameworks & Libraries	React.js, Node.js, Express, Flask, Streamlit, TensorFlow, Keras, Scikit-learn, OpenCV
Cloud & Database	MongoDB, Firebase, Google Cloud Platform (GCP), RESTful API Development
Core Competencies	System Design & Architecture, Full-Stack Development, Deep Learning, Data Analysis
Tools & Hardware	Git, Docker, Linux (Bash), VS Code, Raspberry Pi 5, ESP32, Arduino, Unity3D
Soft Skills	Problem Solving, Team Collaboration, Technical Documentation, Analytical Thinking

PROFESSIONAL EXPERIENCE

Titan Company Limited - Software Engineer Intern

June 2025 – July 2025

- Architected a scalable full-stack system for enterprise HR operations using the **MERN stack**, enhancing internal data processing efficiency by **25%**.
- Secured sensitive infrastructure by implementing **JWT-based authentication** and robust RESTful APIs for **500+ active directory users**.
- Optimized frontend performance by **30%** through React-based dashboards featuring advanced state management and data-driven server-side filtering.
- Engineered integrated **attendance and project management modules** with secure file handling, reducing manual administrative overhead by **40%**.

TECHNICAL PROJECTS

Trinetra Rover – Multi-Modal AI Surveillance Platform

August 2025 – Present

- Spearheaded a semi-autonomous rover that secured **external research funding**, using **Raspberry Pi 5** for edge AI and **ESP32** for sensor fusion.
- Deployed an **Acoustic Classification** model and a **YOLO Visual AI** pipeline, achieving **92% classification accuracy** in identifying threats.
- Developed a high-performance real-time dashboard using **React.js** and **Firebase**, achieving **sub-200ms latency** via UDP command protocols.

ML in VLSI – GNN-based Netlist Reverse Engineering

January 2025 – May 2025

- Applied **Graph Neural Networks (GNNs)** to gate-level netlists to identify circuit sub-blocks and logic functionality with high precision.
- Built a pipeline to convert netlists into **graph structures**, leveraging node features to achieve high-accuracy structural recognition.

Digital IT Helpdesk & Service System

October 2024 – December 2024

- Engineered a centralized IT platform that digitized technical support, achieving a **35% reduction** in ticket resolution turnaround time.
- Integrated workflow-driven tracking for technician assignments and **real-time performance analytics** for administrative oversight.

FreshAudit – Farming Supply Chain & IoT Platform

November 2024

- Built a platform connecting farmers directly to customers, integrating **IoT sensors** for real-time monitoring of agricultural produce quality.
- Programmed an **AI/ML model** to predict crop shelf-life from sensor-derived statistics, targeting a **15% reduction** in supply chain waste.

LEADERSHIP & ACHIEVEMENTS

- Web Development Lead, GDG (Google Developer Group) IIIT Sri City** (2025 – Present): Directing technical mentorship for **750+ students** in modern web stacks and AI-driven application deployment through workshops and bootcamps.
- Electroforge Hackathon Winner** (November 2024): Secured **1st place** among 50+ teams for a real-time IoT water-quality monitoring system with a React-based analytics dashboard.
- Core Sponsorship Team – Abhisarga '25** (Annual Cultural Fest): Negotiated corporate partnerships, contributing to a **20% increase** in the flagship event's operational budget.